Basic electronic engineering

UNIT I – DC Circuits

- Electrical circuit elements:
  - Resistor (R)
  - Inductor (L)
  - Capacitor (C)
- Voltage sources and current sources
- Kirchhoff's laws:
  - Kirchhoff's Current Law (KCL)
  - Kirchhoff's Voltage Law (KVL)
- Analysis of simple DC circuits with DC excitation
- Network theorems:
  - Superposition theorem
  - Thevenin's theorem
  - Norton's theorem

UNIT II – AC Circuits

- Representation of sinusoidal waveforms
- Peak value and RMS value
- Phasor representation
- Power in AC circuits:
  - Real (active) power
  - Reactive power
  - Apparent power
  - Power factor
- Analysis of single-phase AC circuits (series only):
  - R circuit
  - L circuit
  - C circuit
  - RL circuit
  - RC circuit
  - RLC circuit
- Three-phase balanced circuits
- Star (Y) connection:
  - Voltage and current relations
- Delta (Δ) connection:
  - Voltage and current relations

UNIT III – Transformers & Three-Phase Induction Motors

Transformers

- Electromagnetic induction
- Faraday's laws of electromagnetic induction

- Statically induced EMF
- Lenz's law
- B–H characteristics
- Ideal transformer
- Practical transformer
- Transformer losses
- Efficiency of transformer
- Auto-transformer
- Three-phase transformer connections

Three-Phase Induction Motors

- Rotating magnetic field:
    - Generation and concept
- Construction of three-phase induction motor
- Working principle of three-phase induction motor
- Types of induction motors:
    - Squirrel cage induction motor
    - Slip-ring induction motor
- Applications of three-phase induction motors

UNIT IV – Single-Phase Induction Motor & DC Machines

Single-Phase Induction Motor

- Construction
- Principle of operation
- Capacitor start motor
- Capacitor run motor
- Applications

DC Generators

- Dynamically induced EMF
- Fleming's right-hand rule
- Fleming's left-hand rule
- Construction of DC generator
- Principle of operation of DC generator
- EMF equation of DC generator
- Types of DC generators
- Open Circuit Characteristic (OCC)
- Applications of DC generators

DC Motors

- Principle of operation of DC motor
- Types of DC motors
- Applications of DC motors

UNIT V – Electrical Installations

LT Switchgear Components

- Switch Fuse Unit (SFU)
- Miniature Circuit Breaker (MCB)
- Earth Leakage Circuit Breaker (ELCB)
- Molded Case Circuit Breaker (MCCB)

Wiring & Safety

- Types of wires
- Types of cables
- Earthing

Batteries

- Types of batteries
- Important characteristics of batteries

Basic Electrical Calculations

- Energy consumption calculations
- Power factor improvement
- Battery backup calculations